Identification of Environmental Categories for Markovian Deterioration Models of Bridge Decks

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3 authors:

George Morcous
University of Nebraska at Lincoln

Zoubir Lounis
National Research Council Canada

Saeed M. Mirza
McGill University

Identification of Environmental Categories for Markovian Deterioration Models of Bridge Decks

G. Morcous[1]; Z. Lounis[2]; and M. S. Mirza[3]

Abstract: In general, state-of-the-art bridge management systems have adopted Markov-chain models to predict the future condition of bridge elements and networks in different environments when various maintenance actions are implemented. However, the categories used to describe the various possible environments for a bridge element are neither accurately defined nor explicitly linked to the external factors affecting the element deterioration. In this paper, a new approach is proposed to provide transportation agencies with an effective decision support tool to identify the categories that best define the environmental and operational conditions specific to their bridge structures. This approach is based on genetic algorithms to determine the combinations of deterioration parameters that best fit each environmental category. The proposed approach is applied to develop Markovian deterioration models for concrete bridge decks using actual data obtained from the Ministére des Transports du Québec. This application illustrates the ability of the proposed approach to correlate the definition of environmental categories to parameters, such as highway class, region, average daily traffic, and percentage of truck traffic, in an accurate and efficient manner.

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Introduction

Since the early 1980's, transportation agencies have started using bridge management systems (BMSs) to assist them in making systematic and cost-effective decisions regarding the maintenance, repair, and rehabilitation (MRR) of their bridge networks. This is basically due to the aging of highway bridges, increase of operational loads, and severity of environmental impacts, which have led to continuous and extensive deterioration of bridge structures. Moreover, limited funds have made satisfying the bridge MRR needs a challenging task. The primary objective of an effective BMS is to optimize the allocation of the available maintenance funds efficiently to ensure the safety, serviceability, and functionality of bridge structures and networks. The outcomes of the "*1999 Conditions and Performance Report*" issued by the U.S. Department of Transportation (U.S. DOT) showed the effectiveness of BMSs in achieving this objective. This report revealed a steady decline in the percentage of deficient bridges (structurally deficient or functionally obsolete) from 34.6% in 1992 to 29.6% in 1998, the period during which several BMSs have been implemented (U.S. DOT 1999).

[1]Assistant Professor, School of Engineering, Acadia Univ., Wolfville NS, Canada B4P 2R6. E-mail: george.morcous@acadiau.ca

[2]Research Officer, Institute for Research in Construction, National Research Council of Canada, Ottawa ON, Canada K1A 0R6. E-mail: zoubir.lounis@nrc.ca

[3]Professor, Dept. of Civil Engineering and Applied Mechanics, McGill Univ., 817 Sherbrooke W., Montreal PQ, Canada H3A 2K6. E-mail: saeed.mirza@mcgill.ca

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Reliable bridge deterioration models that predict the future condition of bridge elements, systems, and networks are indispensable components of any BMS (American Association of State Highway and Transportation Officials 1993). These models are critical to the assessment of the life cycle costs of the various maintenance strategies and the selection of the optimum strategies based on the total cost accrued over the entire life of the bridge (Hudson et al. 1987). The literature of bridge deterioration models is comprised of several methods that can be categorized into deterministic, stochastic, and artificial intelligence models (Morcous et al. 2002a). The Markov chain is the most widely used stochastic approach for modeling the deterioration of highway bridges and other infrastructure facilities such as pavements, sewer pipes, and water pipes (Madanat et al. 1997; Micevski et al. 2002). Markov-chain models are based on the concept of probabilistic cumulative damage, which predicts changes of element condition over multiple transition periods. The main advantages of these models are: (1) the ability to capture the uncertainty from different sources such as, uncertainty in initial condition, uncertainty in applied stresses, presence of condition assessment errors, and inherent uncertainty of the deterioration process (Lounis 2000), (2) incremental models that account for the current condition in predicting the future condition (Bogdanoff 1978), and (3) applicability to both components and large size networks because of their computational efficiency and simplicity of use (Madanat et al. 1995).

However, Markov-chain models adopted in the state-of-the-art BMSs, such as Pontis (Golabi and Shepard 1997) and BRIDGIT (Hawk 1995), have some drawbacks that affect their applicability and may influence the reliability of their output. For a given bridge element and a specific maintenance action, four Markov-chain models corresponding to four different environmental categories (benign, low, moderate, and severe) are used (Hawk 1995; Thompson et al. 1998). Each category represents a different level of impact of external factors (e.g., traffic, freeze–thaw cycles, and use of deicing chemicals) on the element performance over time.

The objective of these categories is to account indirectly for the influence of these factors on the element deterioration rate and to differentiate between the deterioration of similar elements subjected to dissimilar environments. However, the classification of a bridge element into any of these categories is mostly *ad hoc* and based only on engineering judgment. This subjective classification may result in overestimating or underestimating the element environmental condition, which may lead to assuming either a higher or lower deterioration rate than the actual rate. This in turn will lead to nonoptimal maintenance decisions and inadequate allocation of funds. Furthermore, the absence of an explicit link between the definition of the environmental categories and the external deterioration factors does not allow for an automatic updating of the element category to reflect the changes that may occur in the element environment causing a variation in its deterioration rate.

In this paper, a new approach is proposed to provide transportation agencies with an effective decision support tool in redefining the environmental categories associated with different bridge elements adopted by existing Markov-chain models. This is achieved by linking these categories explicitly to the external deterioration parameters specific to bridge elements and networks. This approach is based on the application of a genetic algorithm (GA) optimization technique to determine the combinations of deterioration parameters that best fit each category. Actual data obtained from the Ministère des Transports du Québec (MTQ) are used to illustrate the use of this approach in developing Markov-chain models for concrete bridge decks for the four environmental categories defined by the state-of-the art BMSs. The flexibility of the new approach, its capability of incorporating additional parameters, compatibility with existing BMSs, and the ease of updating when new data are collected, are its main advantages.

The paper presents the environmental categories used by the existing BMS's in developing Markov-chain deterioration models, the development of Markovian deterioration models for concrete bridge decks using the MTQ data, and illustrates the capability of GAs in identifying environmental categories associated with concrete bridge decks.

Markovian Deterioration Models in Current Bridge Management Systems

A stochastic process is considered as a Markov process if the probability of a future state in the process depends only on the current state and not on how it was attained (Parzen 1962). This property can be expressed for a discrete parameter stochastic process (X_t) with a discrete state space as

$$\begin{aligned} P(X_{t+1}=i_{t+1}|X_t=i_t, X_{t-1}=i_{t-1}, \ldots, X_1=i_1, X_0=i_0) \\ = P(X_{i+1}=i_{t+1}|X_t=i_t) \qquad (1) \end{aligned}$$

where i_t = state of the process at time t; and P = conditional probability of a future condition given the present and past conditions.

A Markov chain is a special case of the Markov process whose development can be treated as a series of transitions between certain states. The use of Markov chains in developing bridge deterioration models is based on the assumption that the condition of a bridge element is independent of its condition history despite the fact that bridge deterioration is a nonstationary process (Lounis and Mirza 2001). The validity of this assumption has to be investigated when sufficient condition data become available.

Discrete parameter Markov chains, which define distinct states of condition and discrete transition time intervals, are commonly used for modeling bridge deterioration because they eliminate the computational complexity of continuous condition states and simplify the decision-making process (Bogdanoff 1978; Madanat and Ibrahim 1995). Most of the current Markov-chain models use a discrete condition rating system that ranges from 1 to 5, where state 1 represents the condition of a new undamaged element, state 5 represents a severely deteriorated element, and states 2, 3, and 4 represent intermediate levels of damage (Lipkus 1994; Thompson et al. 1998). These models predict the probability that a given bridge element in a given environment and a certain initial condition will continue to remain in its current condition state, or change to another condition state, within a one-year period when a particular action is performed. Four different environmental states: Benign, low, moderate, and severe; and three possible maintenance actions: Do nothing, rehabilitate, and replace are defined for each bridge element. For each environmental state and maintenance action, the performance of a bridge element can be predicted once the probabilities of transition from one condition state to another are known. Fig. 1 shows a typical transition probability matrix (P) of order (5×5) for a deteriorating element with the ''do nothing'' maintenance option. If other maintenance options are considered, different transition probability matrices should be used. For the present case, all of the elements of this matrix are 0 except for the diagonal line and the line above it assuming that a bridge element can transit by, at most, one condition state in a year. Given the initial condition vector (P_0) of the bridge element, the future condition vector (P_T) at time (T) can be obtained as follows (Parzen 1962):

$$P_T = P_0 \cdot P^T \qquad (2)$$

Transition probabilities are usually obtained using an expert judgment elicitation procedure, which requires the participation of several experienced bridge engineers (Thompson and Shepard 1994). The outcomes of this procedure are manipulated to generate transition probability matrices for agencies with inadequate condition data. A statistical updating of these probabilities can be undertaken using the Bayesian approach when a statistically significant number of consistent and complete sets of condition data become available over the years (Golabi and Shepard 1997).

Fig. 2 shows an example of four transition probability matrices for concrete bridge decks protected with an asphalt concrete (AC) overlay and subjected to the four environmental categories. Concrete bridge decks are selected because they are considered to be the weakest links and most expensive elements of bridge systems in North America due to corrosion-induced degradation, effect of freezing and thawing cycles, and direct exposure to traffic loads. The probabilities shown in Fig. 2 represent the change of bridge deck condition under normal operational conditions, excluding any emergencies, such as earthquakes, floods, fire, or accidents, when the do nothing maintenance option is adopted. These tran-

$$\begin{array}{c|ccccc|} & 1 & 2 & 3 & 4 & 5 \\ 1 & p_{11} & 1\text{-}p_{11} & 0 & 0 & 0 \\ 2 & 0 & p_{22} & 1\text{-}p_{22} & 0 & 0 \\ 3 & 0 & 0 & p_{33} & 1\text{-}p_{33} & 0 \\ 4 & 0 & 0 & 0 & p_{44} & 1\text{-}p_{44} \\ 5 & 0 & 0 & 0 & 0 & 1 \end{array}$$

Fig. 1. Typical transition probability matrix for the ''do nothing'' maintenance option

	1	2	3	4	5
1	0.98	0.02	0	0	0
2	0	0.97	0.03	0	0
3	0	0	0.97	0.03	0
4	0	0	0	0.96	0.04
5	0	0	0	0	1

Benign environment

	1	2	3	4	5
1	0.95	0.05	0	0	0
2	0	0.94	0.06	0	0
3	0	0	0.94	0.06	0
4	0	0	0	0.92	0.08
5	0	0	0	0	1

Low environment

	1	2	3	4	5
1	0.93	0.07	0	0	0
2	0	0.92	0.08	0	0
3	0	0	0.91	0.09	0
4	0	0	0	0.90	0.10
5	0	0	0	0	1

Moderate environment

	1	2	3	4	5
1	0.87	0.13	0	0	0
2	0	0.86	0.14	0	0
3	0	0	0.85	0.15	0
4	0	0	0	0.83	0.17
5	0	0	0	0	1

Severe environment

Fig. 2. Transition probability matrices of concrete bridge decks with asphalt concrete overlay for the four environments

sition probabilities are derived from extensive literature of bridge deck deterioration and engineering judgment to represent a sample of the Markov-chain models adopted by the existing BMSs.

To demonstrate the differences among the four environmental categories with respect to the predicted performance and service life of concrete bridge decks, the transition probability matrices shown in Fig. 2 are used to develop the deterioration curves in Fig. 3. Each of these curves represents the relationship between the average condition rating of concrete bridge decks and their age for a specific environmental category. If the state 3 is adopted as the minimum acceptable deck condition, the predicted average service lives of bridge decks in benign, low, moderate, and severe environments are 83, 38, 27, and 15 years, respectively. This significant variation in the service life of bridge decks illustrates the considerable impact of the environment on the performance of bridge structures. As a result, it is crucial to develop a rational approach for the accurate classification of bridge elements, systems, and networks in the appropriate environmental categories.

Development of Markovian Deterioration Models for Concrete Bridge Decks

The bridge data used for developing Markov-chain deterioration models are obtained from the MTQ database, which is part of a

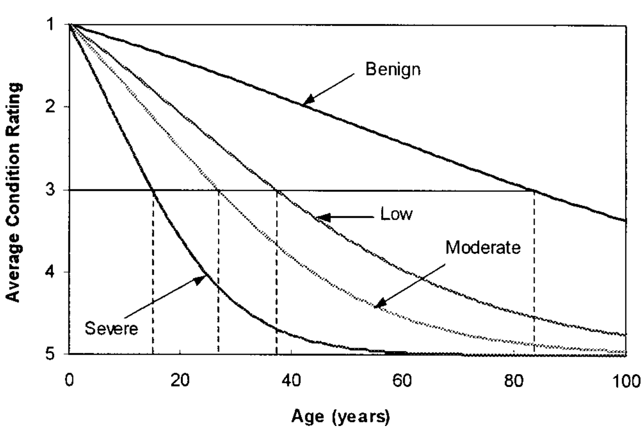

Fig. 3. Deterioration curves of concrete bridge decks with asphalt concrete overlay for different environments

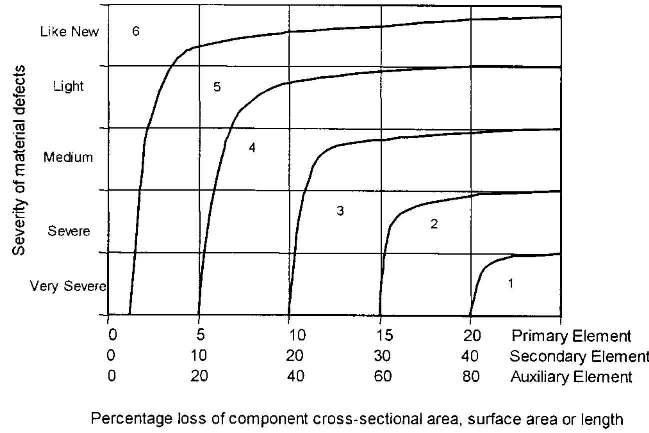

Fig. 4. Ministère des Transports du Québec material condition rating system (MTQ 1995)

comprehensive system for managing the various highway structures in Québec. This system consists of four main modules for identification of maintenance needs, scheduling of rehabilitation strategies, prioritization, and bidding. The database accounts for 9,678 provincially owned highway structures that are grouped into eight categories: culverts, slab bridges, beam bridges, box-girder bridges, truss bridges, arch bridges, cable bridges, and other structures. This database includes three types of data for each highway structure:

1. **Inventory data**, which consist of administrative data (e.g. identification, location, jurisdiction, etc.), technical data (e.g. environment, traffic, postings, etc.), and descriptive data (e.g. geometry, materials, structural system, etc.) (MTQ 1997);
2. **Condition data**, which contain the results of the detailed visual inspections carried out an approximately one-third of the structures every year. This includes: (1) Material condition rating (MCR), which represents the condition of an element based upon the severity and extent of observed defects and (2) performance condition rating (PCR), which describes the condition of an element based upon its ability to perform its intended function in the structure (MTQ 1995). Both the MCR and PCR are represented in an ordinal rating scale that ranges from 1 to 6, where 6 represents the condition of a new and undamaged structure. Fig. 4 shows how the MCR of an element in a highway structure in Québec is determined given the type of element (i.e., primary, secondary, or auxiliary), percentage of material defects in element cross-sectional area, surface area, or length, and severity of these defects (i.e. very low, low, medium, severe, and very severe). Deck and wearing surfaces are examples of primary elements, sidewalks and curbs are examples of secondary elements, and drainage ditches and gutters are examples of auxiliary elements; and
3. **Maintenance data**, which represent the estimated cost and expected time of maintenance/rehabilitation actions recommended as future activities. Data about previous actions are not currently available in the MTQ database.

To obtain a consistent, complete, and adequate data set for developing Markov-chain deterioration models for concrete bridge decks, the data in the MTQ database are screened by filtering out the records with missing condition data or incomplete inventory data. The screening process resulted in 9,181 bridge spans in 3,440 beam bridges. The deck of each span consists of

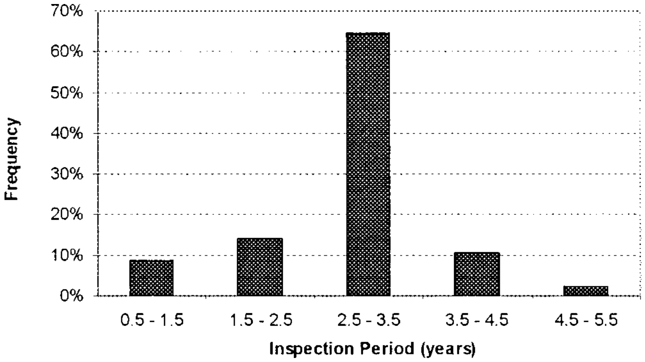

Fig. 5. Histogram of inspection periods in Ministére des Transports du Québec data

seven elements that are evaluated in every inspection: wearing surface, drainage system, two exterior faces, two end portions (each portion has a length equal to twice the girder depth), and the middle portion. The overall condition of the bridge deck (MCR) is calculated as the aggregation of the MCRs of the seven elements using element weights obtained by expert judgment based on their relative importance. Since the aggregated MCR is a cardinal number, the deterioration models are developed using the MCRs of the elements (i.e., ordinal numbers) to allow the use of discrete parameter Markov chains.

Literature on Markov-chain models includes different methods for generating transition probability matrices from the condition data. Examples are the percentage prediction (Jiang et al. 1988), regression-based optimization (Butt et al. 1987), Poisson distribution and negative binomial models (Madanat and Ibrahim 1995), and ordered probit and random effects models (Madanat et al. 1997). The regression-based optimization method is the most-widely used approach for estimating these matrices for bridge and other facilities (e.g., pavements and sewers) (Bulusu and Sinha 1997). This method uses a nonlinear optimization function to minimize the sum of absolute differences between the regression curve that best fits the condition data and the conditions predicted using the adopted Markov-chain model (Madanat and Ibrahim 1995).

Because the regression model used in this method is significantly affected by any prior maintenance actions for which the data are not readily available in the MTQ database, the percentage prediction method is used. In this method, the probability p_{ij} of transition in a bridge condition from state i to state j can be estimated using the following equation (Jiang et al. 1988):

$$p_{ij}=n_{ij}/n_i \tag{3}$$

where n_{ij} = number of bridge elements that deteriorate from state i to state j within one transition period; and n_i = total number of bridge elements in state i before the transition.

The use of this method requires at least two consecutive condition rating records for a large number of bridge elements at different condition states, which is the case of the MTQ database. Since the condition data used in developing these probabilities have variable inspection periods as shown in Fig. 5, the transition period for the matrix is 2.8 years, which is the average of inspection periods. Fig. 6 shows the 2.8 years transition probability matrix developed using the percentage prediction method for concrete bridge decks protected with an AC overlay when the do nothing maintenance alternative is implemented.

	6	5	4	3	2	1
6	0.65	0.32	0.03	0.00	0.00	0.00
5	0.00	0.88	0.09	0.02	0.01	0.00
4	0.00	0.00	0.85	0.10	0.03	0.01
3	0.00	0.00	0.00	0.88	0.08	0.04
2	0.00	0.00	0.00	0.00	0.93	0.07
1	0.00	0.00	0.00	0.00	0.00	1.00

Fig. 6. Transition probability matrix of concrete bridge decks with asphalt concrete overlay and 2.8 years transition

In order to compare the transition probabilities estimated from the literature review and engineering judgment (Fig. 2) with those developed based on the "actual" condition data (Fig. 6), a standard condition rating system has to be used. The MTQ rating system was selected as the standard rating system, while the conditions described using the 1 to 5 rating system were converted using linear interpolation. Fig. 7 shows the deterioration curve generated using the MTQ data and the curves presented earlier in Fig. 3 after conversion. Fig. 7 indicates that the transition probability matrices developed based on literature and engineering judgment have reasonable accuracy for network level analysis since the deterioration curve developed using the field data represents the average of the deterioration curves that correspond to the four environmental categories.

A subset of the MTQ data was also used to validate the state independence assumption of Markov-chain models using a simple frequency test and inference test. The results of these tests showed that this assumption is not totally realistic, however, it may be considered acceptable for network level analysis (Morcous et al. 2002b).

Identification of Environmental Categories Using Genetic Algorithms

In order to provide BMS users with a rational approach to identifying the environmental categories of concrete bridge decks and selecting the most appropriate Markov-chain models, the inventory and condition data stored in the BMS database are utilized to correlate the definition of these categories with the governing deterioration parameters. The approach will overcome the arbitrary nature of the current procedures used in categorizing bridge

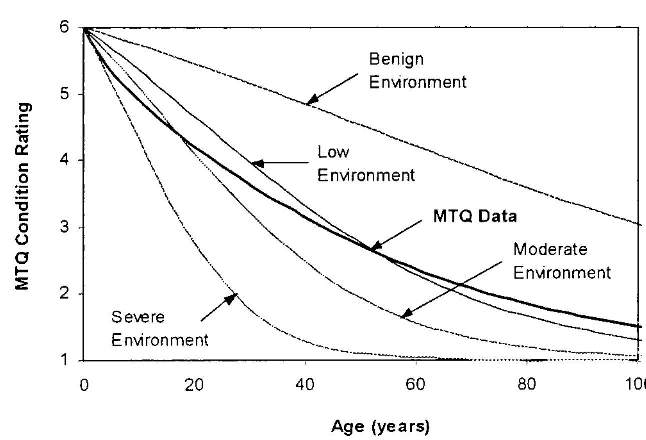

Fig. 7. Comparison of deterioration curves of concrete bridge decks with asphalt concrete overlay

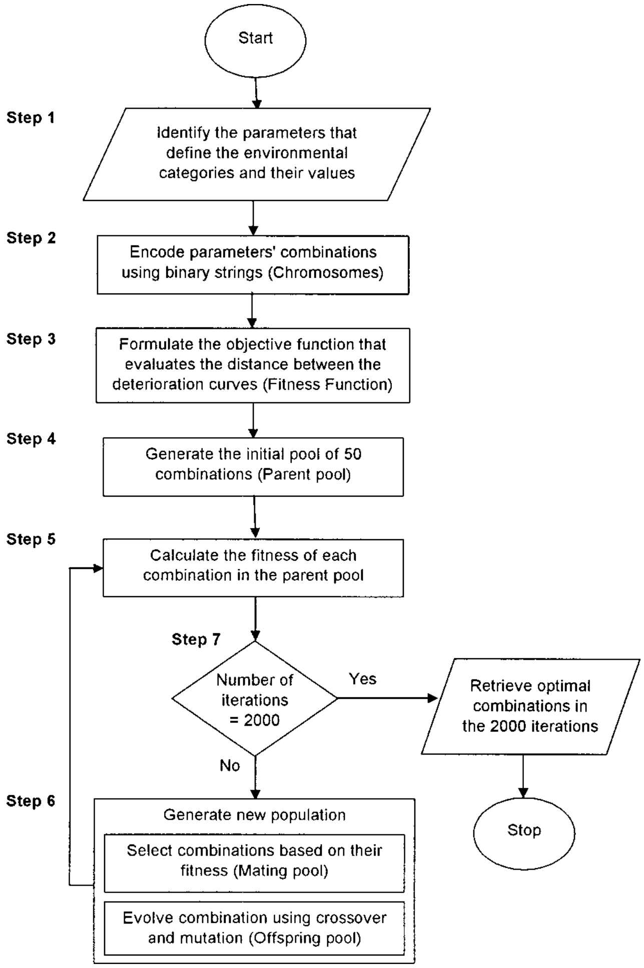

Fig. 8. Flow chart of the problem-solving process in genetic algorithm

decks and allow the continuous updating of the deck category when changes in its environment occur.

The goal of the proposed approach is to find the combinations of deterioration parameters that best define each environmental category and, consequently, select the deterioration model that best represents the actual performance. The determination of these combinations can be formulated as an optimization problem in which the objective function is the minimization of the distance between the deterioration curve of bridge decks subjected to a combination of deterioration parameters and the deterioration curve that represents a specific environmental category. In most engineering optimization problems, the objective function is defined as an explicit function of problem parameters that should be minimized or maximized. However, the objective function in the current problem cannot be defined explicitly in a simple equation that is easy to evaluate but rather it is a complex function that is evaluated in several operations. First, for a given combination of deterioration parameters, the function queries the MTQ database for all bridge decks that satisfy this combination. Second, using the retrieved set of data, the transition probability matrix is generated using the percentage prediction method presented in the section entitled "Development of Markovian Deterioration Models for Concrete Bridge Decks." Third, the average deterioration curve, specific to this set of bridge decks, is determined using the aforementioned matrix. Finally, the distance between the generated deterioration curve for a given combination and the deterioration curve that corresponds to one of the four environmental categories is calculated as the sum of absolute differences. These four operations evaluating the objective function are automated using Microsoft Access queries and Microsoft Excel macros. The complexity of the objective function and its nonexplicit nature make the use of traditional optimization approaches complicated, cumbersome, and time-consuming. Moreover, the computational complexity of the problem increases proportionally with the number of the environmental categories, deterioration parameters, and bridge decks; and exponentially with the number of parameter values, which may result in a computationally prohibitive optimization problem for traditional approaches. Given the nature of the objective function, GA optimization techniques are used to solve this environmental categorization problem.

Table 1. Governing Deterioration Parameters of Concrete Bridge Decks

Deterioration parameter	Value class	Description
Highway (H)	H_1	Express and national
	H_2	Regional and collector
	H_3	Local and others
Region (R)	R_1	Eastern
	R_2	Northern
	R_3	Central
	R_4	Western
Average daily traffic (A)	A_1	$<5{,}000$
	A_2	$\geqslant 5{,}000$
Percentage of truck traffic (T)	T_1	$<10\%$
	T_2	$\geqslant 10\%$

Genetic algorithms are robust, practical, and general-purpose stochastic search-based optimization techniques. They were developed by Holland in the early 1970's based on the principles of natural selection and genetics. GAs are one of the commonly used techniques in evolutionary computation, which is comprised of different methods for solving optimization problems by imitating living beings (Holland 1992; Gen and Cheng 2000). Since the early 1980's, GA applications have been extensively used in a wide range of disciplines, such as optimization, pattern recognition, robotics, and operation control. In infrastructure management, several researchers have adopted GAs to optimize the allocation of maintenance funds for asset networks, such as the selection of optimal maintenance strategy for bridge decks by Liu et al. (1997) and optimization of maintenance and rehabilitation decisions for pavements by Fwa et al. (1996). Moreover, GAs has several advantageous characteristics over traditional optimization techniques (Goldberg 1989). First, for a given optimization problem, GAs adopt the solution pool approach in searching for the optimum solution instead of the single-solution approach of the traditional techniques, implying that the GA can efficiently handle problems with large search spaces. This is necessary for the current problem since the size of the search space increases exponentially with the number of deterioration parameters considered and the possible values of each parameter. Second, for the transition from one pool of solutions to another pool that contains better solutions, GAs use probabilistic rules rather than deterministic rules of traditional techniques. These probabilistic rules increase the likelihood of finding an optimal solution without being

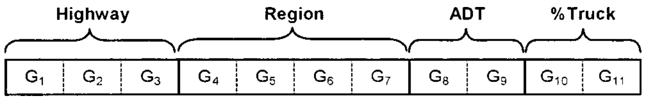

Fig. 9. Genetic coding of problem parameters

trapped in local optima. Third, for evaluating a solution, GAs only use information on the objective function and do not require any information on its gradients, or auxiliary properties, like traditional approaches. This allows the GA to be applied to problems that cannot be easily formulated in a mathematical formulation, which is the case with the current problem.

Problem-Solving Process in Genetic Algorithms

This section presents the steps of applying the problem-solving process in GAs to the aforementioned problem. Fig. 8 shows a simplified flow chart of this process.

1. The first step consists of identifying the parameters that define the four environmental categories used by the current BMSs. These parameters, shown in Table 1, were determined based on the data obtained from the MTQ database. Each parameter has an impact on the deterioration of concrete bridge decks and, therefore, it will be linked to the definition of the environmental categories. Table 1 also shows the classes that identify the values of symbolic parameters and the value groups of numeric parameters. The value groups of numeric parameters (e.g., average daily traffic and percentage of truck traffic) were determined based on literature of bridge deck deterioration. It should be noted that the numbers of environmental categories, deterioration parameters, and value classes may differ from one transportation agency to another. The four parameters used in this application were selected based on their significance and data availability in the MTQ database.
2. The second step in this process is encoding the problem parameters into a genetic representation (i.e., chromosomes). This is because GAs cannot work directly with problem parameters in their original representation. Each chromosome represents a single solution and consists of several genes that can be manipulated by the genetic operations described later. There are several encoding methods that can be used (Goldberg 1989): binary encoding, real-number encoding, integer or literal permutation encoding, and general data structure encoding. Binary encoding, which transfers a solution into a binary string (with 0 and 1 symbols), is the most widely used method because of its simplicity and applicability to many problems. Fig. 9 shows the chromosome structure (number of genes and their arrangement) of a combination of parameters using the binary encoding method. This structure can be expanded when additional parameters or values classes become available.
3. The third step is the formulation of the objective function, also termed the fitness function, which has been discussed in detail in the section "Identification of Environmental Categories Using Genetic Algorithms."
4. The fourth step of the problem-solving process in GAs is to randomly select an initial population of solutions and improve it iteratively in several generations. This population constitutes the initial pool of solutions (parent pool) which has a predefined size N ($N=50$ in this application).
5. The fifth step is to calculate the fitness of each solution in this parent pool using the objective function defined in the third step.
6. In the sixth step, if the population does not satisfy a predefined stopping criterion, two processes are implemented to generate a new population, namely selection and evolution processes. Selection is considered the driving force of the GA since its selects the most promising chromosomes from the parent pool and generates a mating pool that has the same number of chromosomes (N). Two selection schemes can be used: Roulette wheel or tournament scheme (Chong and Zak 2001). The former scheme, which used in this example, is a stochastic procedure that determines the probability of selection for each chromosome proportional to its fitness value. Evolution is the second process in which the chromosomes of the mating pool are manipulated by the two genetic operations, crossover and mutation, to produce the offspring pool as shown in Fig. 10. In the crossover operation, pairs of parent chromosomes are selected randomly with probability p_c (0.5 in this example) and each pair exchanges the genes of the two chromosomes at a crossing site selected randomly. The objective of this operation is to produce offspring solutions that have combined features from the parent solutions. In the mutation operation, parent chro-

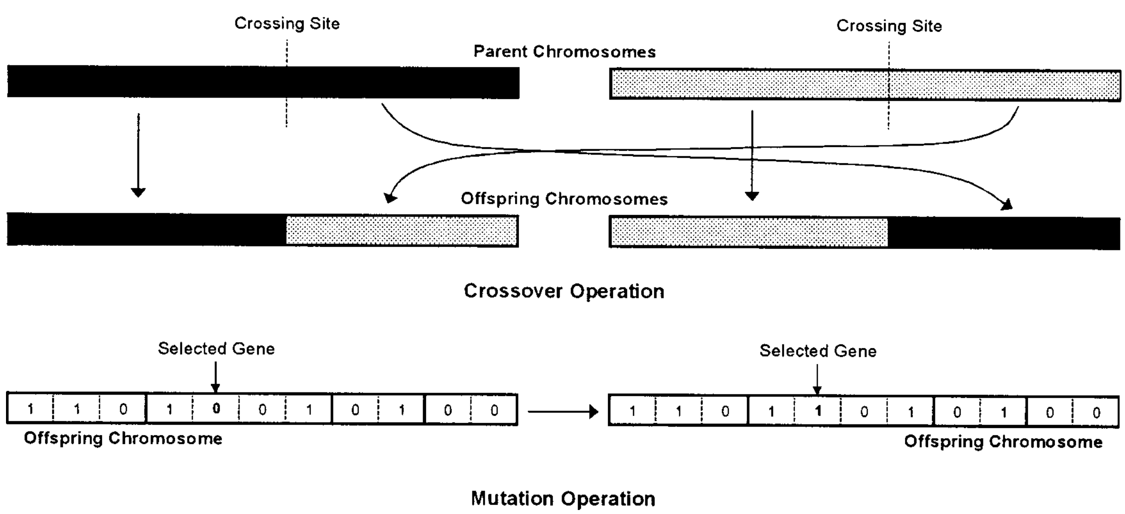

Fig. 10. Genetic algorithm operations: Crossover and mutation

Table 2. Optimal Combinations of Deterioration Parameters for the Four Environmental Categories

	Highway class			Region class				ADT class		Truck class		Fitness values			
Environmental category	H_1	H_2	H_3	R_1	R_2	R_3	R_4	A_1	A_2	T_1	T_2	Benign	Low	Moderate	Severe
Benign	1	0	0	1	0	0	0	0	1	1	0	2.55	54.13	73.91	98.47
Low	0	1	1	0	1	0	1	0	1	1	0	53.12	1.87	18.89	43.45
Moderate	1	1	1	0	0	1	0	0	1	0	1	77.91	25.68	6.36	18.78
Severe	1	0	0	0	0	1	0	1	0	0	1	91.14	38.91	19.13	7.17

mosomes are selected and their genes are changed randomly from 0 to 1, or vice versa, and with probability p_m (0.01, in this example). The objective of this operation is to introduce random changes into a small fraction of solutions to try potential solutions that have never been selected. The resulting population is then evaluated using the objective function and used as a new parent population. The fifth and sixth steps are repeated iteratively until the stopping criterion is satisfied. The GA tool, called *EVOLVER 4.0* (Palisade Company 2000), was used to implement the selection and evolution processes in this example.

7. The seventh step of the problem-solving process in GAs is to define a suitable stopping criterion for the previous iteration. In this application, the performance of the genetic algorithm is monitored using the progress graph that plots the number of iterations versus the average solution of the population. The progress graph that was developed while searching for the combination of deterioration parameters that best fit each of the four environmental categories has shown that the average solution does not improve after 2,000 iterations. Therefore, the chosen stopping criterion was based only on the maximum number of iterations, which was taken as 2,000 iterations. The solution corresponding to the minimum value of objective function in these 2,000 iterations was considered the optimal solution.

Optimal Solutions Obtained by Genetic Algorithms

The output of the problem-solving process presented earlier is comprised of the four combinations of bridge deck deterioration parameters that best describe the four environmental categories adopted by the current BMSs. These combinations are considered the GA-generated optimal solutions that are shown in their coded form (chromosomes) in Table 2. Each optimal solution can be easily interpreted (i.e., decoded) by finding the value(s) of the deterioration parameters, from Table 1, that correspond to the cells whose values are equal to 1.0 in Table 2. For example, the first row in Table 2 describes the bridge decks classified into the "benign" environmental category as those bridge decks on express and national highways, located in Eastern Quebec, with an average daily traffic greater than 5,000, and a percentage of truck traffic less than 10%. Also, the values of the fitness function corresponding to the four environmental categories are calculated for each optimal solution to illustrate the deviation of this solution from each environmental category.

Fig. 11 shows the deterioration curves generated for the optimal solutions as well as those generated using the existing Markovian deterioration models. Fig. 11 indicates the high level of similarity between the GA-generated deterioration curves and the existing deterioration curves. This illustrates the capability of GAs in finding the combination of the deterioration parameters

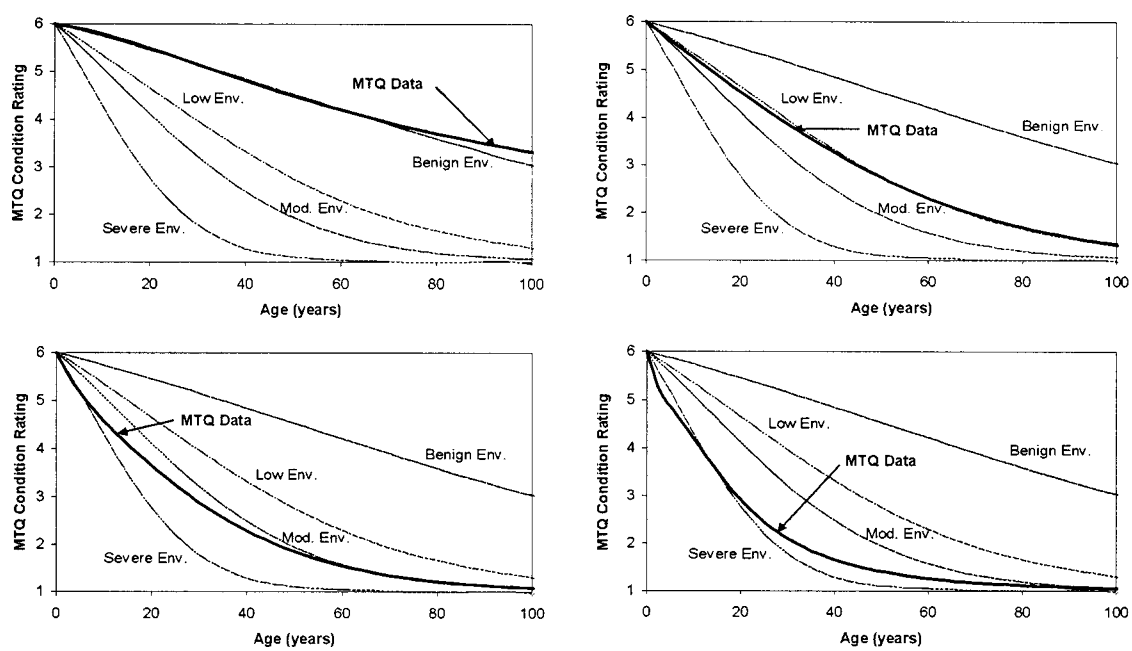

Fig. 11. Genetic algorithm-generated deterioration curves for the four environmental categories

that accurately represent each environmental category.

It should be noted that the four combinations obtained by the GA approach and shown in Table 2 are presented as guidelines for BMS users to correlate the definition of the environmental categories with the governing deterioration parameters. Having a bridge deck with parameter values H_i, R_j, A_k, and T_1 that do not fall into one of the identified categories means that the environment of this deck is in between any two of the identified categories. This is because the four environmental categories were defined as simplification for the different possible environments of concrete bridge decks. Therefore, the user has to approximate the environment of the bridge deck to one of these four categories based on the proposed guidelines.

Summary and Conclusion

Bridge deterioration models constitute an integral part of any BMS because they predict the future condition of bridge elements and, consequently, their maintenance needs. The stochastic Markov-chain models are used extensively in modeling bridge deterioration because of their ability to capture the time-dependence and uncertainty of the deterioration process in addition to their computational efficiency and reliability especially at the network level. However, the existing Markov-chain models have some drawbacks that may affect the accuracy of their outcomes. These models assume that a bridge element has to be classified into one of the following environmental categories: benign, low, moderate, or severe, in order to account for the impact of the governing deterioration parameters. These categories are not well-defined and the classification of a bridge element to any category is made in a subjective and *ad hoc* manner, which may result in either overestimating or underestimating the element deterioration. Also, this classification does not allow the automatic updating of the element deterioration rate when any changes occur in the element environmental and operational conditions.

The goal of the proposed approach is to find the combinations of deterioration parameters that best define each environmental category and, consequently, select the deterioration model that best represents the actual performance. This approach is based on using the bridge inventory and inspection data stored in the BMS database in conjunction with GA optimization techniques to find the optimal combinations of deterioration parameters. GAs were selected because of the complexity and nonexplicit nature of the objective function. The data obtained from the MTQ was used to define the four environmental categories for Markov-chain models of concrete bridge decks based on four parameters: highway class, region, average daily traffic, and percentage of truck traffic. This illustrative example showed the feasibility and capability of the proposed approach in correlating the four environmental categories with these parameters in an accurate and efficient manner.

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